Final-Term Report for the BDM Capstone Project

Submitted by

Name: Avnish Shivpure

Roll number: 22ds3000023

Email ID: 22ds3000023@ds.study.iitm.ac.in

Title: Inventory Management Optimization and Profit Maximization for Pramod Traders - A Data-Driven Approach

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1. Executive Summary and Title

The BDM project titled “Inventory Management Optimization and Profit Maximization for Pramod Traders - A Data-Driven Approach”, in this report provides detailed information on the results and recommendations used to optimize inventory management and maximize profit for Pramod Traders. The data which is used in the project was collected from Tally (inventory management software). A lot of data was not suitable for the analysis so only data on 620 products was considered.
During the analysis following results and Observations were found:

- Volume Analysis showed a consistent adherence to the Pareto principle. A few products such as Standard switches, PVC plates, insulation tape and machine screws made a significant portion of sales. And revealed that only the top 15 products contribute to 50% of the sales volume.
- Movement Analysis showed that the months of May to July, October, December and January. The analysis identified April as the least productive month in terms of sales volume.
- Weekly Analysis showed a distinctive patterns in inventory movement, that movement is constant throughout the week but on Wednesday and Friday it exhibits the high inventory movement, whereas Monday sees the lowest. This information presents opportunities to strategically align operations and promotions with the high and low movement days.
- Monthly Sales Analysis showed that months from September to December stand out as the highest revenue months. While April and August were marked as low sales volume and revenue andless business-friendly months.
- Weekly Sales Analysis gave a characteristic bell curve, with Thursday at its center.

- Category-wise Analysis of Inventory and Revenue showed that Electrical products exhibit consistent high sales throughout the year. Hardware, on the other hand, showed peak sales in May and July. Sanitary products, specifically PVC, cpvc, and CP fitting, experience peak sales in April and November, although they generate lower revenue.

Interpretation of Results and Recommendations on the basis of analysis:

- Suggestion would be to optimize sales volume and focus on top-performing products. Regularly stock high-volume items to prevent stock outs.
- Maximize revenue by promoting high-margin electrical products through establishing networks with industry professionals to identify opportunities for expansion and partnerships.
- Plan inventory management based on monthly and weekly sales trends. Concentrate efforts and promotional activities during peak revenue generating months, such as June and December.
- Expand the electrical product range and promote sanitary products during peak months and consider bundling these with hardware products to maximize revenue.

2. Detailed Explanation of Analysis Process/Method

The analysis will be done in the following ways:

- **Cleaning of data**

1. Empty columns like party information item number item description will be removed.
2. Columns which were unnecessary will also be removed like product alias, godown, item batch.
3. Data with incorrect values like opening and closing stock for stock summary which was not accurate will also be removed.

4. Data entry for purchase rate in sales data was not accurate for some items and was also reversed(i.e. purchase rate became selling rate) such data will also be removed.

- **Removing Errors**

1. The error regarding purchase rate was removed by using VLOOKUP function with the help of purchase data as it was accurate.
2. Not all of the items were purchased during the period of 10 months so some rows will have empty purchase rates which will require the input of the owner for the correction, any error occurring with the VLOOKUP function will be removed and replaced with '0' using IFERROR function.

- **New Changes:** Out of all the data the following were removed
    a. 933 products which didn't have any data.
    b. 868 products which had inventory in inventory data but had zero recorded sales.

These changes left only 620 products which had a proper record of sales. So, in this project the findings were based on these products only.

- **<u>All the previous data was used with:</u>**
  The data was distributed in previous mid-term files was used, now the following files were also used :

1. **Stock Up:** Stock Up
   This data has the list of loosely bought products

2. **Top 20%:** Top 20%
   This data has list of best products in terms of sales and volume
3. **Total Stock Out:** Total Stock Out
   This data has a list of products which don't have proper stock information or were out of stock.

- **Method of Analysis for Different Problems**

1. **Problem 1**: Finding the most suitable product on the basis of number of units sold and the amount of revenue generated for the item:

    1.1. For this problem the Item List data will be useful as it can easily provide the Pareto chart for items in each category which will help in determining the most profitable and the most sold product in that particular category.

    1.2. Good products can also be identified with the help of normalization of values with a particular weight.

2. **Problem 2**: Inventory Management Issues

    2.1. This problem can be solved with the help of list data as it can provide information on fast moving and slow moving products.

    2.2. With the help of a Scatter plot between quantity of product and the margin of individual item we can easily understand which product is fast moving with high margin or low margin and slow moving items with high or low margin.

    2.3. Stock out product list (category wise) can also be found with the Item List Data by simply subtracting purchased quantity from sold quantity and the more the negative value we get the more Stockout that particular item is facing.

3. **Problem 3**: Information on Weekly Basis

    3.1. With the Date-wise Data analysis trend line and chart done to identify the most b of the week and also to identify the best day to stock up, this can be simply done by plotting a Trend line chart where x-axis can be taken as weeks while each Trend line can represent a month.

    3.2. Here the weekly division for each month will be done with the help of a Pivot table where months will be taken as rows and week days will be taken as columns and in the value section we can put either average margin or number of units sold.

4. **Problem 4**: Finding list of data with errors

4.1. As discussed above during the data analysis process it was found that there were some errors with the purchasing stock and the opening stock data.

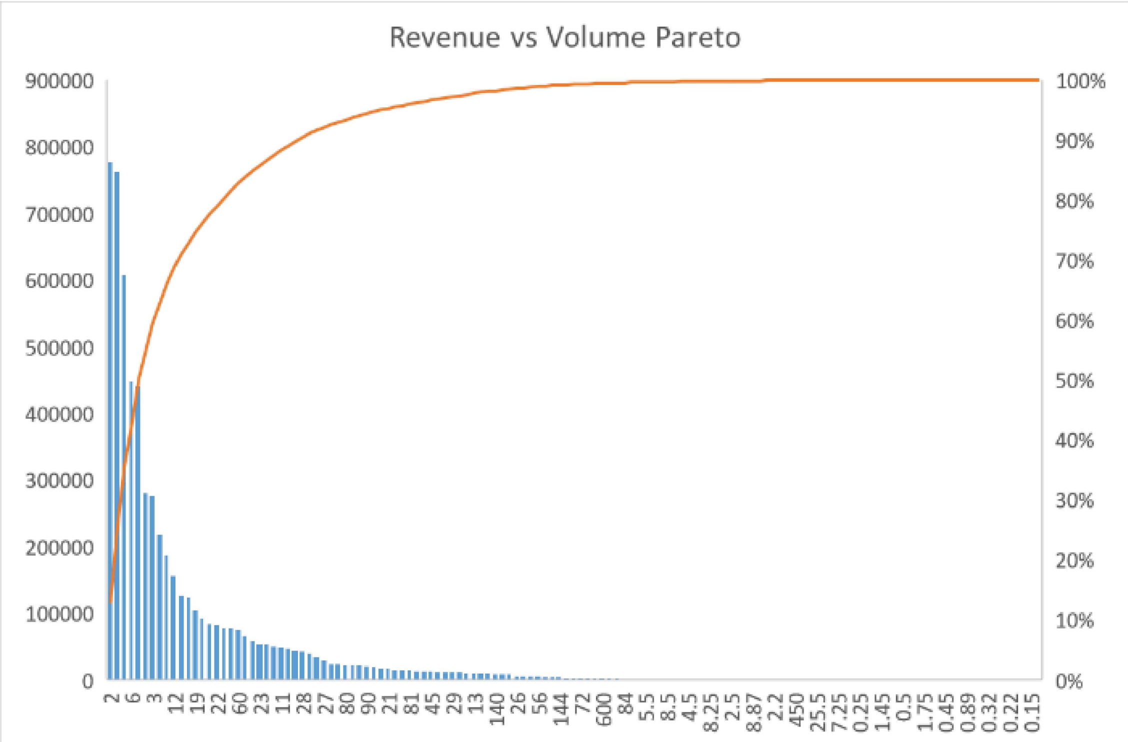
4.2. Upon discussion with the shop owner regarding the data error it was discovered that software Tally is not considering previous years inventory and previous inventory data while providing any output regarding inventory data, so the owner also requested for the list of such items which were facing errors.

This problem could be solved with the item list data simply filtering the data category wise where the filter is set to "Smallest to Largest", with this whenever we get zero or negative value in purchase quantity column we can interpret it as an error of the software and the list product name and the column of purchasing quantity can be given to the owner.

3. <u>Results and Findings</u>

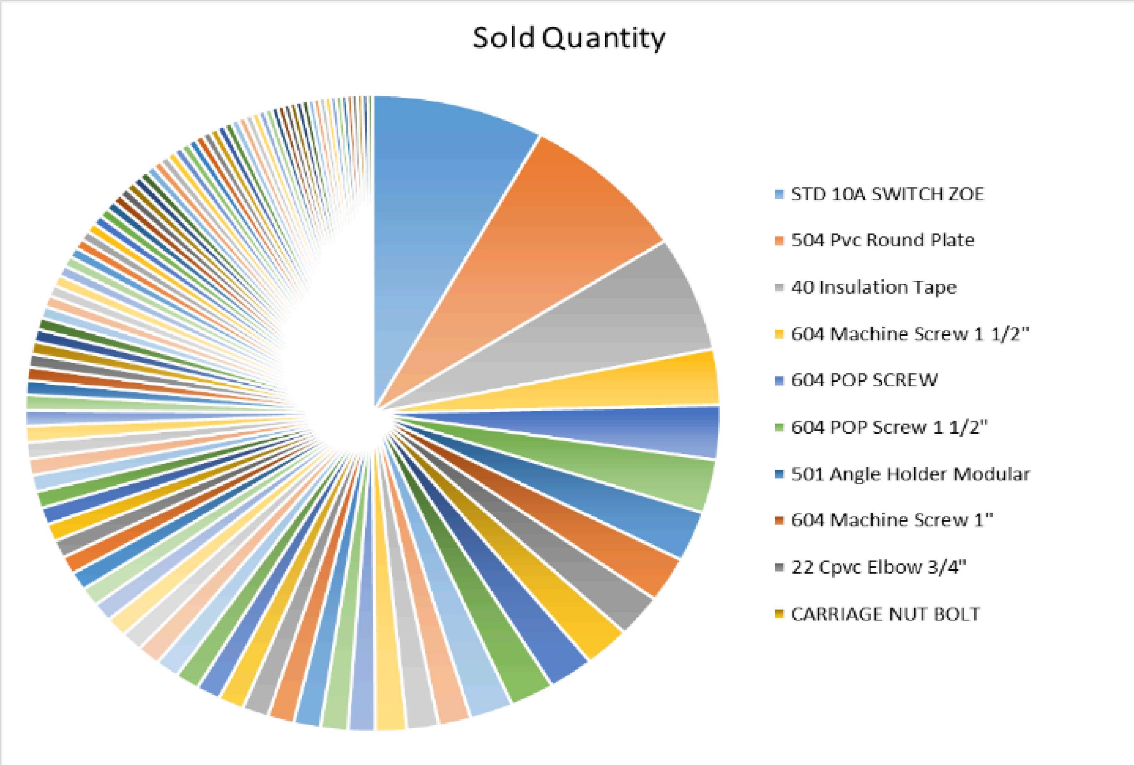
3.1. Volume analysis:

3.1.1. Volume analysis results the unit sold by the shop followed a Pareto principle throughout the unit sold by the shop; this can be seen in the Pareto distribution of the units sold.

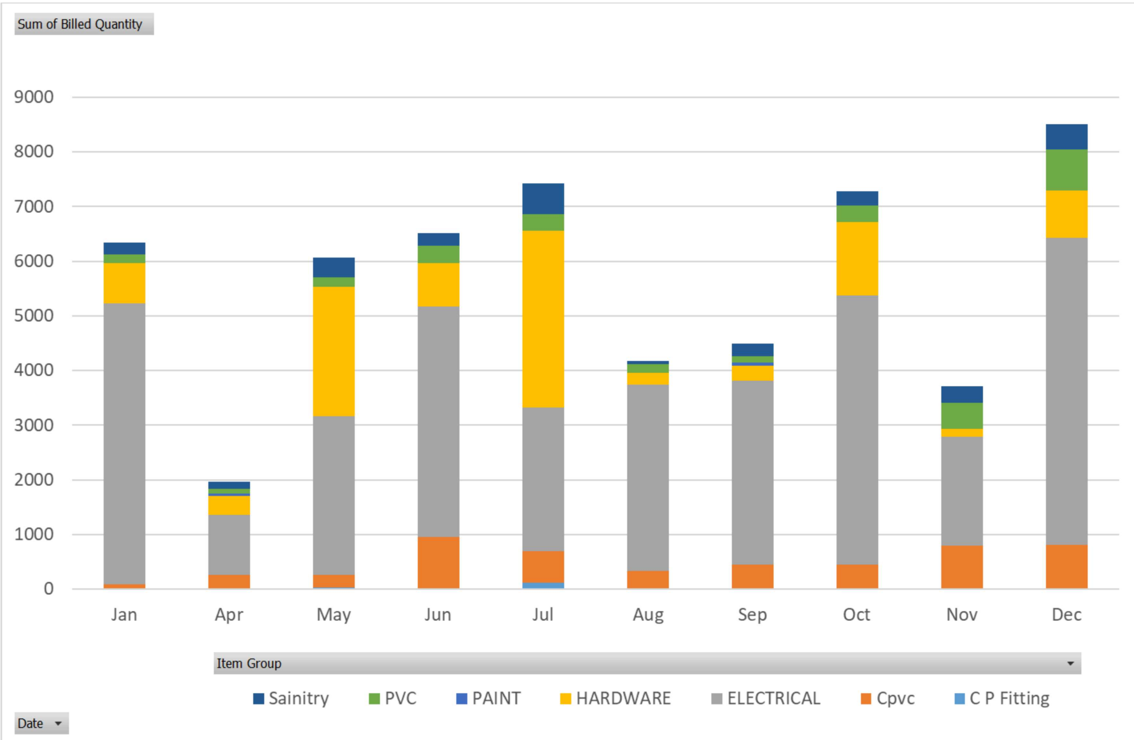


3.1.1.1. Here, only a few products like standard switches, PVC plates, insulation tape, machine screws, etc can be seen.

3.1.2. The following pie chart of most selling 100 products provides further confirmation for that and only top 15 products are able to generate 50% of the sales volume.

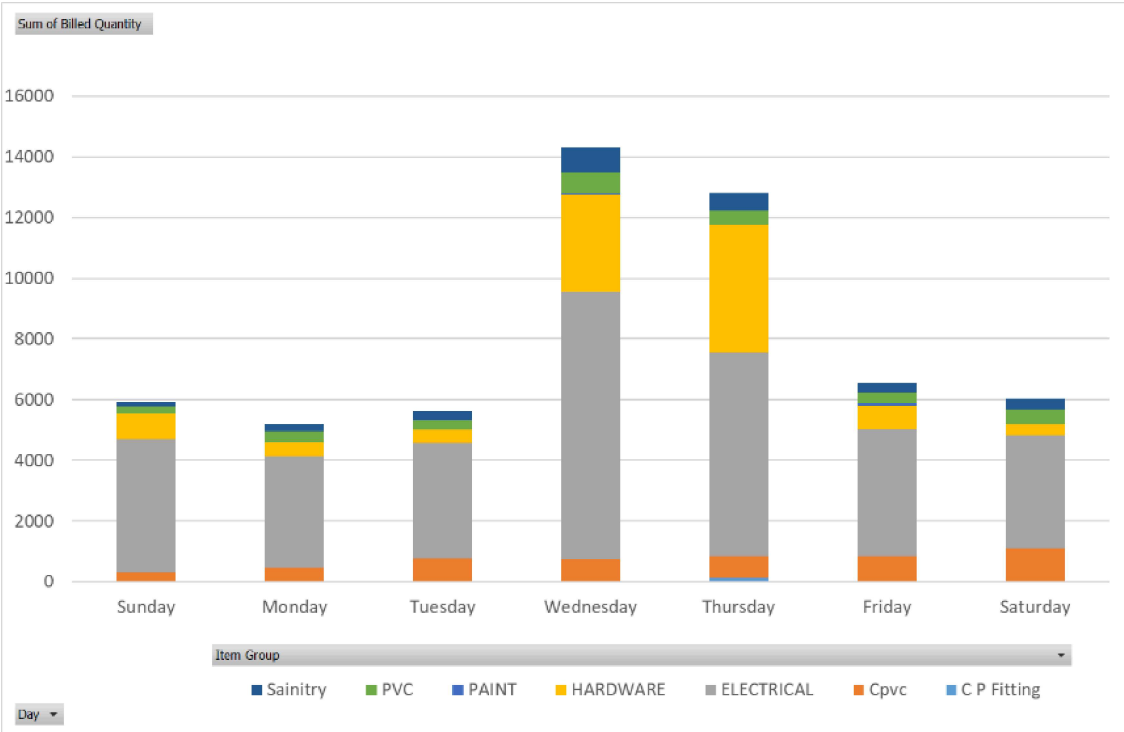


3.1.3. Movement analysis of the sold quantity gives information about the best selling month.



3.1.3.1. Here, the best volume generating period was from month of main and from October to January(except November) and April was the least generating month

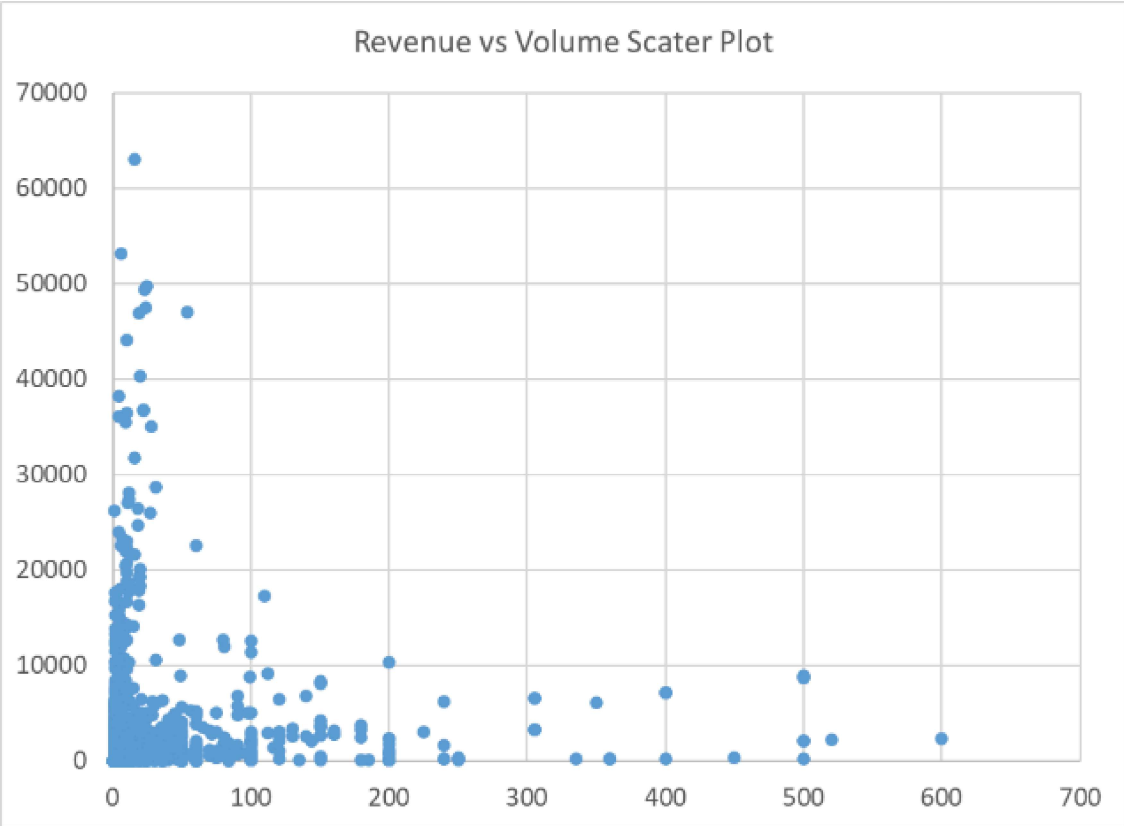
3.1.4. Weekly analysis of sold quantity gives information about the days of week which had the highest movement which had the lowest or highest movement.



3.1.4.1. Here, we found that Wednesday and Friday showed the highest inventory movement Monday had the least.

3.2. Item analysis:

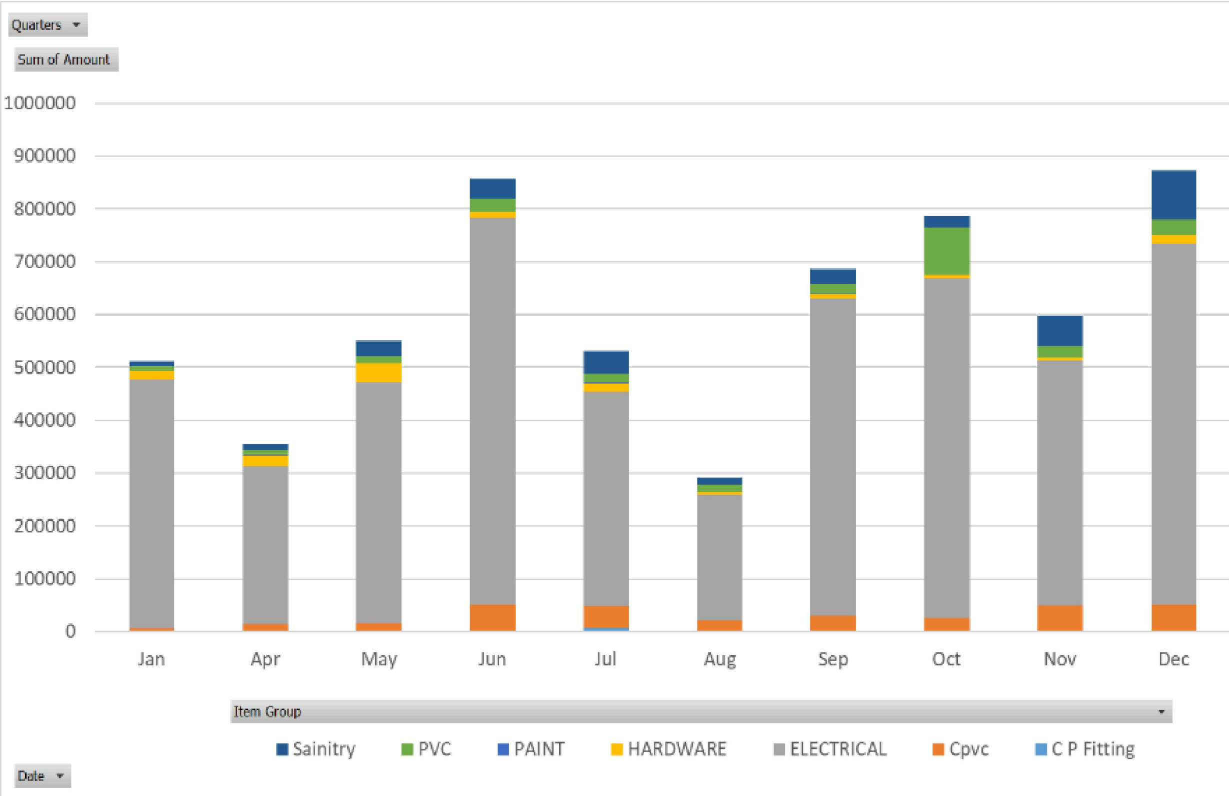
3.2.1. A list of 15 most selling and 15 most revenue generating products was formed through the Scatter plot of units sold and revenue generator as shown below.



3.2.2. It was found that between most selling and most revenue generating product the product of electrical category a similarity and they also dominated the revenue generated products which tells us that the product of electrical category are most sold and most revenue generating product including high margin low volume product.

3.3. Monthly sales analysis:

The monthly sales chart gave information about the sales of products for each category in each month.



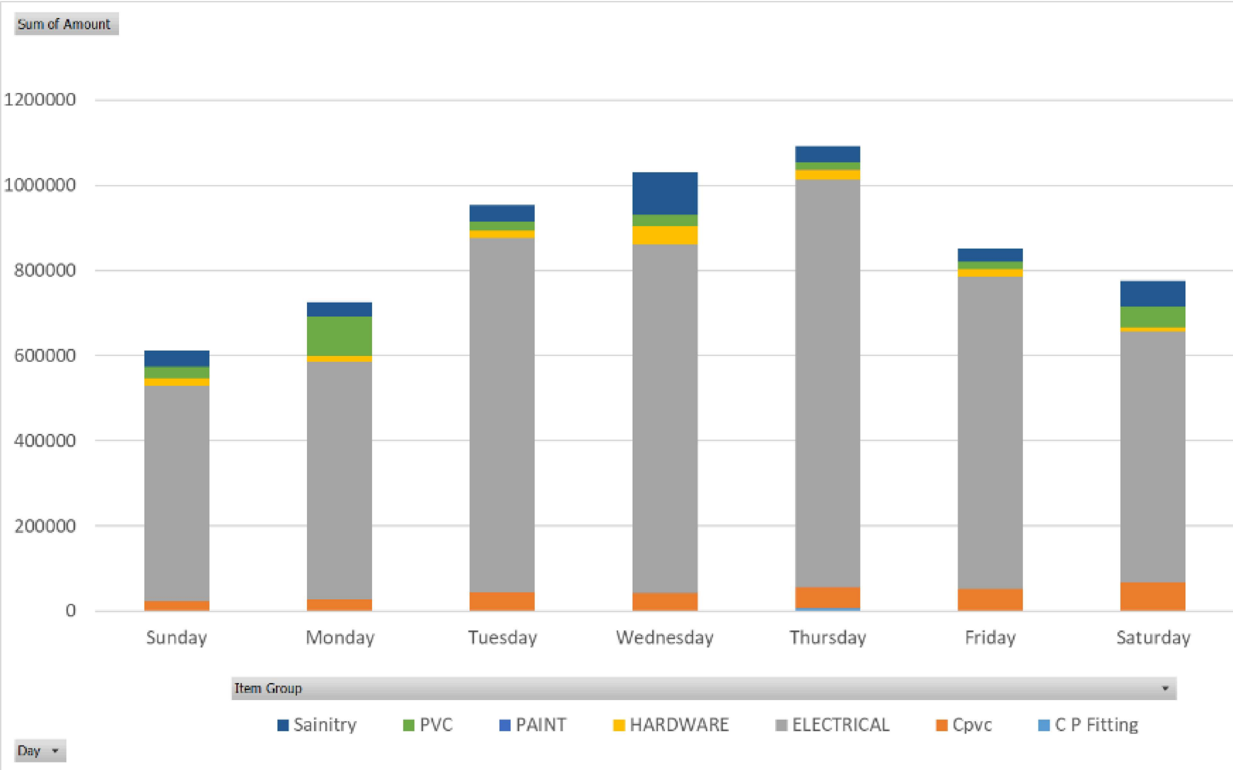
3.3.1. It showed that the months of January and December generated the most revenue and the period of September to December provided the best overall sales.

3.3.2. For months of January, May and July which had high sales volume Got Low revenue generation.

3.3.3. Months of April and August had low sales volume and revenue, which suggest that they are less business friendly months.

3.4. Weekly sales analysis:

The weekly sales analysis of the shop follows a normal distribution with Thursday at the centre, as it is seen in the graph.

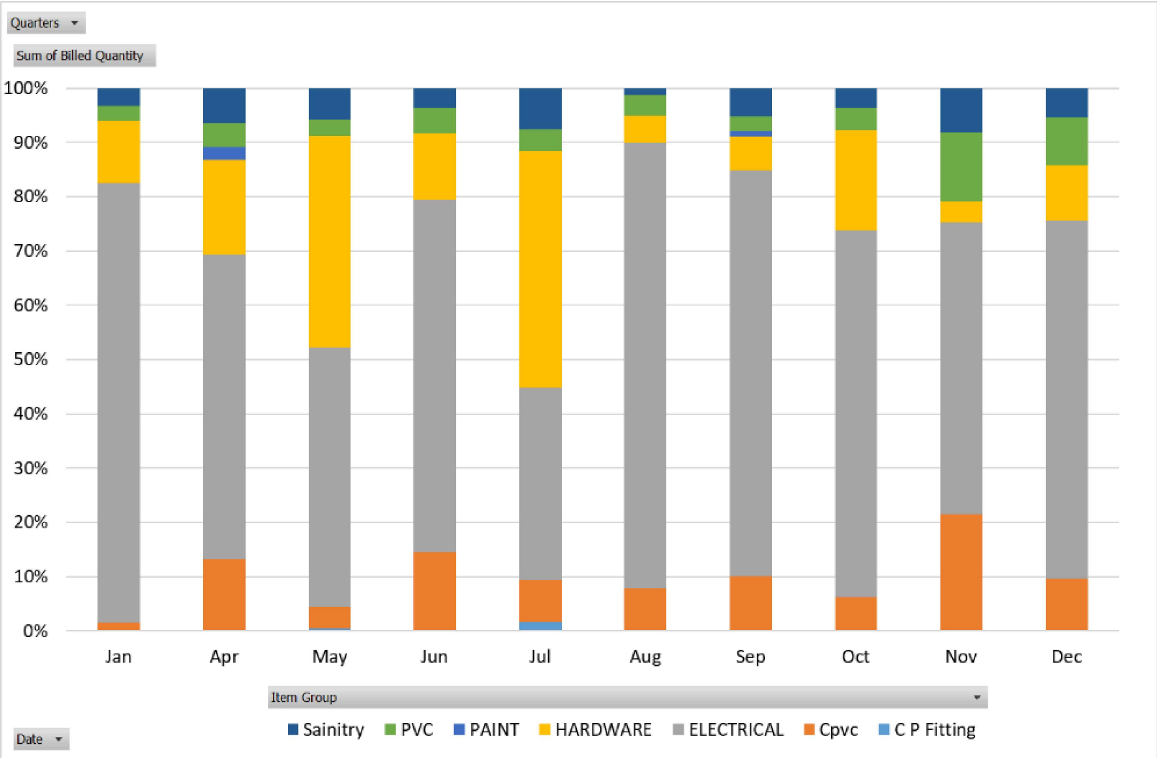


3.4.1. From the graph we can understand that from Tuesday to Wednesday is the period where most revenue is generated and Sunday has the least.

3.4.2. While the number of the units sold throughout the week is constant except Wednesday and Thursday(which also had good sales), Tuesday and Friday generated high revenue with less sales volume.

3.5. Category wise Analysis of Inventory and Revenue:

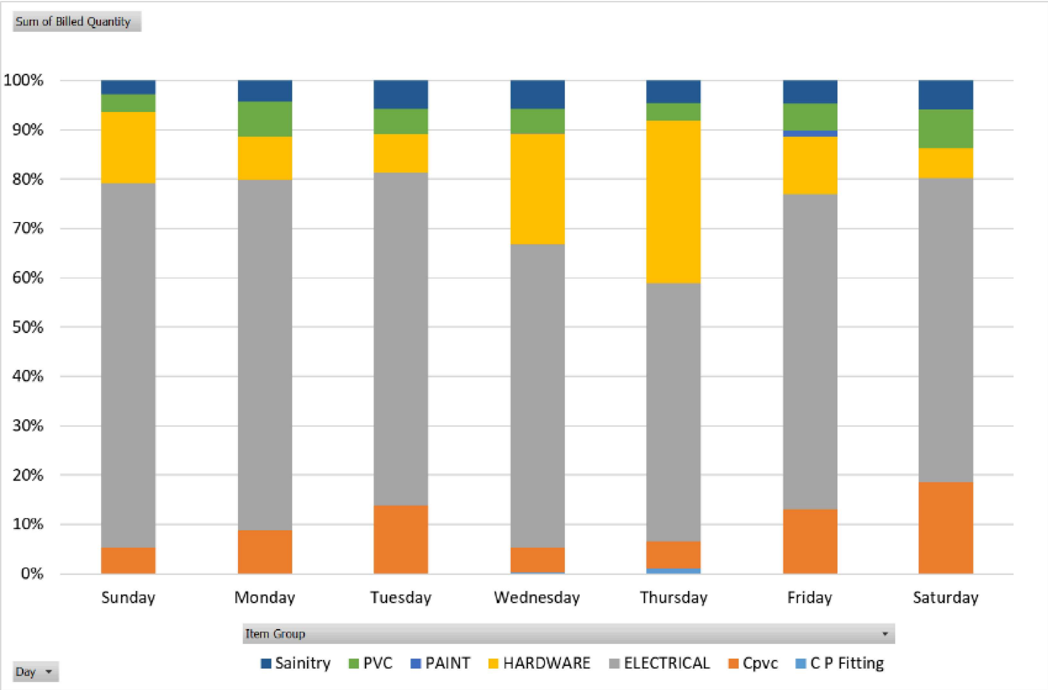
3.5.1. Inventory analysis on a monthly basis showed that electrical products are the most sold category throughout the year.



3.5.1.1. And Hardware was most sold during the month of May and July

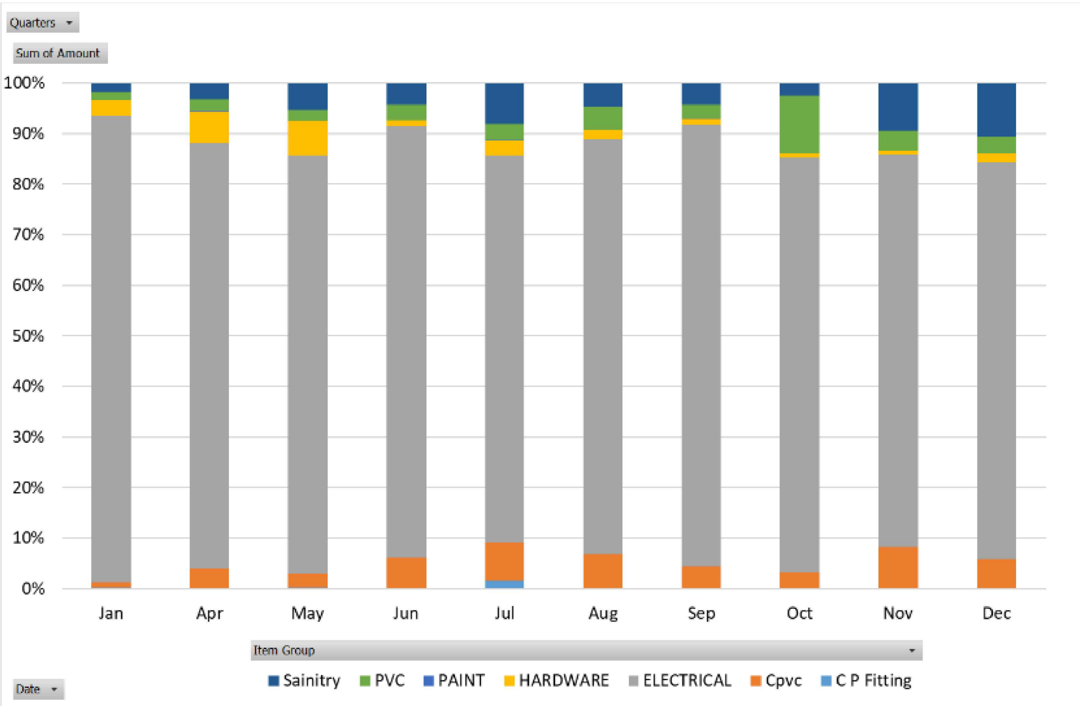
3.5.1.2. Sanitary products like PVC, cpvc, CP fitting world most sold product in the month of April and November(which also were low revenue months).

3.5.2. Inventory analysis on weekly basis showed that the electrical goods had throughout the week.



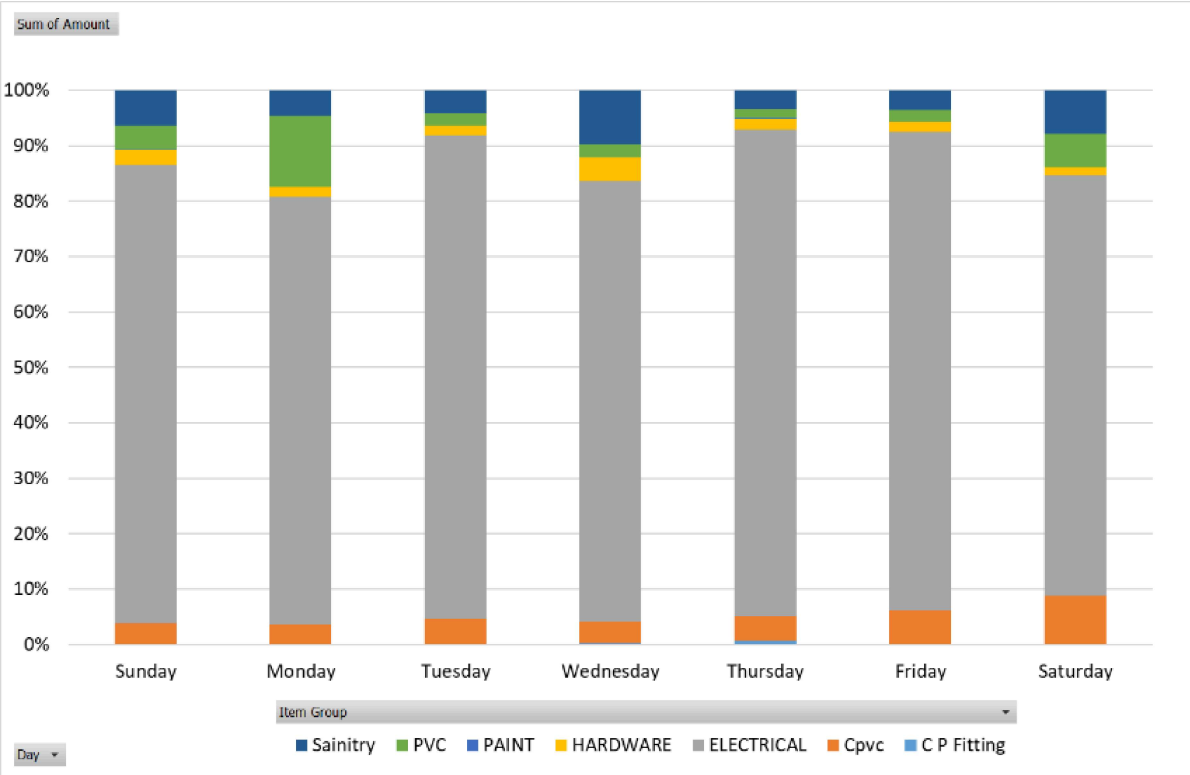
Hardware related goods were mostly sold on the day of Wednesday and Thursday.

3.5.3. Sales analysis on a monthly basis showed that the most of the revenue was earned by the electrical category.



PVC, CP fitting and cpvc had good sales in month of October to December and July.

3.5.4. Sales analysis on weekly basis showed that the electrical goods were sold most revenue generating as usual.



While sanitary goods(including PVC and cpvc) generating good revenue on Monday, Wednesday and Saturday.

4. <u>Interpretation of Results and Recommendation</u>

4.1. Interpretation of Volume analysis:

4.1.1. I would suggest to Focus on Top Performers (as given in Top 20%): And concentrate efforts on products which follow the Pareto principle, such as standard switches, PVC plates, insulation tape, and machine screws. These items generate a significant portion of the shop's sales volume.

4.1.2. Stock up the frequently sold products regularly on a monthly or weekly basis

4.2. Interpretation of Item analysis:

4.2.1. Utilize High-Margin Electrical Products: I would recommend most selling and most revenue-generating products(as collected in Top 20%), particularly within the electrical category.

4.2.2. Then I would recommend strategically promoting high-margin electrical products for increased revenue by either networking with Builders, Architects and Contractors to gain information of new constructions and customers.

4.3. Interpretation of Monthly sales analysis:

4.3.1. I would suggest <u>*Seasonal Planning*</u> to optimize inventory management based on monthly sales trends.

4.3.2. Stock up before high-revenue months like July and December, and adjust inventory levels during slower months like April and August. <u>Use discounts</u> and promotions more during these specific months to attract customers.

4.4. Interpretation of Weekly sales analysis:

4.4.1. From Tuesday to Friday are high revenue days, So I would suggest not stock-up on these days.

4.4.2. Strategic Promotion Days: Capitalize on higher revenue generation on Tuesdays and Fridays by running special promotions or discounts to encourage higher sales volumes(because they generate high revenue even with low volume), to increase the overall revenue on these days.

4.5. Interpretation of Category wise Analysis of Inventory and Revenue:

4.5.1. For each category:

4.5.1.1. Maximize **Electrical Category**: It is clear that electrical products dominate revenue, then consider expanding offerings within this category. By introducing new electrical products and organizing the store layout to guide customers toward these high-revenue items.Also arrange high demand products in the outer region of the shop.

4.5.1.2. Promote **Sanitary Products**: To improve sales in specific months, such as April and November, create targeted promotional offers or bundles for sanitary products like PVC, cpvc and CP Fittings. Manage the required stock in months of less demand as these products occupy large space.

4.5.1.3. For **Hardware Product**: These products are of loose category and have specific demands, so I would suggest bundling these products.

4.6. ***Optimizing Marketing:*** I would suggest focusing marketing efforts more intensively on Wednesdays and Thursday, when inventory movement is higher. This can lead to increased sales and revenue during these days and also increase customer awareness for the shop.

4.7. Reduce the sales of Paints by keeping it on order only as they are the least selling products.